GM - 2023

Q6432108
Question Paper SL No.

2023

MATHEMATICS

[English Version]

3 Hours 15 Minutes

(First 15 minutes for reading the question paper)

Full Marks — { 90 — **For Regular Candidates**
100 — **For External Candidates**

Special credit will be given for answers which are brief and to the point.
Marks will be deducted for spelling mistakes, untidiness & bad handwriting.

[The answers of the question Nos. 1, 2, 3 are to be written at the beginning of the answer script mentioning the question numbers in the serial order. Necessary calculation and drawing must be given on the right hand side by drawing margins on the first few pages on the answer-script. Tables and Calculators of any type are not allowed. Approximate value of π may be taken as $\frac{22}{7}$, if necessary. Graph paper will be supplied, if required. Arithmetic problems may be solved by algebraic method.]

[Alternative questions of 11 are given for sightless candidates on page No. 15]

[Additional Question number 16 is only for external candidates on Page No. 16]

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1. Choose the correct option in each case from the following questions: $1 \times 6 = 6$

(i) Three friends A, B, C started a business with Capitals Rs. x, $2x$ and y respectively, at the end of term profit is Rs. z, then the share of profit of A is

(a) Rs. $\frac{xz}{3x+y}$ (b) Rs. $\frac{2xz}{3x+y}$

(c) Rs. $\frac{z}{2x+y}$ (d) Rs. $\frac{xyz}{3x+y}$

(ii) Number of solutions of the equation $x^2 = x$ are

(a) 1 (b) 2

(c) 0 (d) 3

(iii) If two circles touch each other internally, then the number of common tangents of the circles is

(a) 1 (b) 2

(c) 3 (d) 4

(iv) For any value of θ the maximum value of $5 + 4\sin\theta$ is

(a) 5 (b) 1

(c) 9 (d) 4

(v) If the ratio of the volumes of two solid spheres is $27 : 8$, then the ratio of their curved surface area is

(a) $1 : 2$ (b) $9 : 4$

(c) $1 : 8$ (d) $1 : 16$

(vi) Three values of a variable are 4, 5 and 7, if their frequencies are $p - 2$, $p + 1$ and $p - 1$ respectively and Mean is 5.4, then value of p is:

(a) 1 (b) 2

(c) 3 (d) 4

2. Fill in the blanks (any **five**): $1 \times 5 = 5$

(i) If the amount of Rs. 180 after one year be Rs. 198, then the rate of simple interest is

.....

(ii) If mean proportional of (a^2bc) and $(4bc)$ is x, then the value of x is .....

(iii) If $\tan\theta\cos 60^\circ = \frac{\sqrt{3}}{2}$, then the value of $\sin(\theta - 15^\circ)$ is .....

(iv) If $\angle A$ and $\angle B$ are complementary then $\angle A + \angle B =$ .....

(v) The median of the numbers 8, 15, 10, 11, 7, 9, 11, 13 and 16 is .....

(vi) The shape of a pencil with one end sharpened is the combination of a ..... and a

.....

3. Write True or False (any **five**): $1 \times 5 = 5$

(i) In compound interest if the rate of interest in first three years is $r_1\%$, $r_2\%$, $2r_3\%$ respectively, then the amount for the principal P at the end of three years is

$P\left(1+\frac{r_1}{100}\right)\left(1+\frac{r_2}{100}\right)\left(1+\frac{r_3}{100}\right)$.

(ii) The values of $\cos 36^\circ$ and $\sin 54^\circ$ are equal.

(iii) One tangent can be drawn on a circle from an external point.

(iv) The compound ratio of $2ab : c^2$, $bc : a^2$ and $ca : b^2$ is $1 : 1$.

(v) If the numerical values of curved surface area and volume of a sphere are equal, then radius will be 3 units.

(vi) The Mode of the data 1, 2, 4, 3, 5, 2, 3, 2, 5, 2 is 2.

4. Answer the following questions (any **ten**): $2 \times 10 = 20$

(i) Find the rate of simple interest per annum when the interest of some money in 5 years will be $\frac{2}{5}$ part of its principal.

(ii) In a business capitals of A and B are in the ratio $\frac{1}{7} : \frac{1}{4}$. If they make a profit of Rs. 11,000 at the end of the year, calculate the share of their profit.

(iii) If the sum of the roots of the equation $x^2 - x = k(2x - 1)$ is 2, then find the value of k.

(iv) If $b \propto a^3$ and a increases in the ratio of $2 : 3$, then find in what ratio b will be increased.

(v) AB and CD are two chords of a circle. If we extend BA and DC, they intersect each other at the point P. Prove that $\angle PCB = \angle PAD$.

(vi) In $\triangle ABC$, L and M are two points on the sides AC and BC respectively such that $LM \parallel AB$ and $AL = (x - 2)$ unit, $AC = 2x + 1$ unit, $BM = (x - 3)$ unit and $BC = 2x$ unit. Determine the value of x.

(vii) Two circles touch each other externally at the point C. A direct common tangent AB touches the two circles at the points A and B. Find the value of $\angle ACB$.

(viii) If $\tan 2A = \cot(A - 30^\circ)$, then find the value of $\sec(A + 20^\circ)$.

(ix) If $\tan\theta = \frac{8}{15}$, find the value of $\sin\theta$.

(x) If the volume of a right circular cone is V cubic unit, base area is A sq. unit and height is h unit, then find the value of $\frac{Ah}{3V}$.

(xi) Find the ratio of the volumes of a solid right circular cylinder and a solid right circular cone of equal radii and equal heights.

(xii) If 6, 8, 10, 12, 13, x are in increasing order and their mean and median are equal, then find the value of x.

5. Answer any **one** question: 5

(i) The number of smokers is decreasing at the rate of $6\frac{1}{4}\%$ per year due to publicity of anti-smoking. If at present number of smokers in a town are 22500, find the number of smokers of that town 2 years ago.

(ii) In a partnership business, the ratio of capitals of three friends is $5 : 4 : 3$. After 4 months 1st friend withdraws his half of the capital and after 8 more months total profit is Rs. 61,050. Find the share of profit of three friends.

6. Answer any **one** question: 3

(i) Solve: $\frac{x-3}{x+3} - \frac{x+3}{x-3} + 6\frac{6}{7} = 0 \quad (x \neq 3, -3)$

(ii) If the price of 1 dozen pens is reduced by Rs. 6, then 3 more pens will be got in Rs. 30. Calculate the price of 1 dozen pens before the reduction of price.

7. Answer any **one** question: 3

(i) If $x = \frac{1}{2-\sqrt{3}}$ and $y = \frac{1}{2+\sqrt{3}}$, then find the value of $\frac{1}{x+1} + \frac{1}{y+1}$.

(ii) If $x \propto y$ and $y \propto z$, then show that $\frac{x^3+y^3+z^3}{3xyz}$ is a constant.

8. Answer any **one** question: 3

(i) If $\frac{a^2}{by+cz} = \frac{b^2}{cz+ax} = \frac{c^2}{ax+by} = 1$, then show that $\frac{1}{1+x} + \frac{1}{1+y} + \frac{1}{1+z} = 1$.

(ii) If the fourth and fifth of the five numbers in continued proportion are 54 and 162 respectively, find the first number.

9. Answer any **one** question: 5

(i) Prove that in a cyclic quadrilateral opposite angles are supplementary.

(ii) Prove that the tangent to a circle at any point on it is perpendicular to the radius that passes through the point of contact.

10. Answer any **one** question: 3

(i) $ABCD$ is a cyclic quadrilateral. Bisectors of $\angle DAB$ and $\angle BCD$ intersect the circle at X and Y respectively. If O be the centre of the circle, find $\angle XOY$.

(ii) Prove that a cyclic trapezium is an isosceles trapezium.

11. Answer any **one** question: 5

(i) Draw a right angled triangle of which two sides containing the right angle have the lengths 5 cm and 6 cm. Now draw the incircle of the triangle.

(ii) Construct a square of equal area of an equilateral triangle of side 7 cm.

12. Answer any **two** questions: $3 \times 2 = 6$

(i) If $\cos\theta = \frac{x}{\sqrt{x^2+y^2}}$, then prove that $x\sin\theta = y\cos\theta$.

(ii) Radius of a circle is 7 cm. Find the angle in radians which subtends by an arc of this circle of length 5.5 cm at the centre of the circle.

(iii) Show that $\frac{\tan\theta+\sec\theta-1}{\tan\theta-\sec\theta+1} = \frac{1+\sin\theta}{\cos\theta}$.

13. Answer any **one** question: 5

(i) Angle of elevation of the top an incomplete tower from a point at a distance 50 m from its foot is 30°. How much should the height of the tower be increased so that the angle of elevation of the top will be 45° from that point?

(ii) From the roof of the building the angle of depression of the top and foot of the lamp post are 30° and 60° respectively. Find the ratio of the heights of building and lamp post.

14. Answer any **two** questions: $4 \times 2 = 8$

(i) Two solid spheres with the radii of 1 cm and 6 cm lengths are melted and a hollow sphere with outer radius 9 cm is made. Determine the inner radius of the new hollow sphere.

(ii) The height of a right circular cone is twice its radius of base. If the height were seven times of the diameter of the base, then volume of the cone would have been 539 cu.cm more. Find the height of the cone.

(iii) Curved surface area of a right circular cylindrical wooden log of uniform density is 440 sq. decimeter. Weight of 1 cubic decimeter of wood is 3 kg and weight of the log is 18.48 quintal. Find the diameter of the log.

15. Answer any **two** questions : $4 \times 2 = 8$

(i) If the arithmetic mean and total frequency of the following distribution are 50 and 120 respectively then find the value of f_1 and f_2:

Class	0 - 20	20 - 40	40 - 60	60 - 80	80 - 100
Frequency	17	f_1	32	f_2	19

(ii) Construct the table of cumulative frequency (greater than type) and draw the ogive from the following frequency distribution :

Class	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60
Frequency	7	10	23	50	6	4

(iii) Find the mode of the following frequency distribution :

Class	50 - 59	60 - 69	70 - 79	80 - 89	90 - 99	100 - 109
Frequency	5	20	40	50	30	6